

Vertex, Axis, Intercepts and Opening Direction

Answer Key

Algebra 1 Parabola Features and Transformations

Quick Answers

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|-----------|--------------|---------------|--------------|
| 1. -4 | 4. -2, 4, -9 | 7. -3, 3, -9 | 10. -1, 5, 9 |
| 2. 5, 1 | 5. -8, -1, 3 | 8. -2, -6, -2 | |
| 3. -9, -5 | 6. 4, -5 | 9. 1, 5, -8 | |
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Curriculum

Level: Algebra 1 Parabola Features and Transformations

HSA-SSE.B.3 – problems 5, 9

HSF-IF.A–HSF-IF.C – problems 1, 2, 3, 4, 6, 7, 8, 10

Difficulty 1–5 of 5 across 20 tagged checks.

Common wrong answers

1. If they got 32: read the sign of h straight off the parentheses, putting the vertex at $x = -3$
3. If they got 7: used $x = b$ over $2a$ instead of minus b over $2a$, landing on $x = 2$
6. If they got -32: dropped the minus sign on a when computing the axis, landing on $x = -3$
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1. $y = (x - 3)^2 - 4$.

Step 1. This is vertex form $y = a(x - h)^2 + k$ with $a = 1$, $h = 3$, $k = -4$. The subtraction inside the parentheses is what makes h positive.

Step 2. Check by substituting the axis value: $y(3) = (3 - 3)^2 - 4 = -4$.

Step 3. Vertex $(3, -4)$, axis of symmetry $x = 3$, opening upward because $a = 1 > 0$.

Sketch. y -intercept $y(0) = 9 - 4 = 5$; by symmetry the curve also passes through $(6, 5)$. Setting $y = 0$ gives $(x - 3)^2 = 4$, so the x -intercepts are 1 and 5 — two units each side of the axis, as symmetry demands.

Common wrong answer: a vertex height of 32 — the sign of h was read straight off the parentheses, putting the vertex at $x = -3$.

2. $y = -(x + 2)^2 + 5$.

Step 1. Rewrite the inside as $(x - (-2))$, so $h = -2$ and $k = 5$; the leading coefficient is $a = -1$.

Step 2. $y(-2) = -(0)^2 + 5 = 5$, and because $a < 0$ this is the highest point: the parabola opens downward.

Step 3. Vertex $\boxed{(-2, 5)}$, axis $x = -2$, opening downward.

Step 4. y -intercept: $y(0) = -(0 + 2)^2 + 5 = -4 + 5 = \boxed{1}$

Sketch. Symmetric partner of the y -intercept is $(-4, 1)$. Setting $y = 0$ gives $(x + 2)^2 = 5$, so the x -intercepts are about -4.2 and 0.2 .

3. $y = x^2 + 4x - 5$.

Step 1. Standard form with $a = 1$, $b = 4$, $c = -5$. Axis of symmetry: $x = -\frac{b}{2a} = -\frac{4}{2} = -2$.

Step 2. Substitute to get the vertex height: $y(-2) = 4 - 8 - 5 = -9$.

Step 3. Vertex $\boxed{(-2, -9)}$, axis $x = -2$, opening upward.

Step 4. The y -intercept is c : $y(0) = \boxed{-5}$

Sketch. Factoring $(x + 5)(x - 1)$ gives x -intercepts -5 and 1 , and they straddle $x = -2$ as they must.

Common wrong answer: $7 - x = \frac{b}{2a}$ was used instead of $-\frac{b}{2a}$, putting the axis at $x = 2$.

4. $y = x^2 - 2x - 8$.

Step 1. The x -intercepts are where $y = 0$, so factor: two numbers with product -8 and sum -2 are -4 and 2 .

Step 2. $x^2 - 2x - 8 = (x - 4)(x + 2)$, so $x - 4 = 0$ or $x + 2 = 0$.

Step 3. x -intercepts $\boxed{x = -2 \text{ and } x = 4}$

Step 4. The axis of symmetry is midway between them: $x = \frac{-2+4}{2} = 1$, and $y(1) = 1 - 2 - 8 = -9$.

Step 5. Vertex $\boxed{(1, -9)}$, opening upward.

Sketch. y -intercept -8 , with symmetric partner $(2, -8)$.

5. $y = 2(x - 1)^2 - 8$.

Step 1. Vertex form with $h = 1$, $k = -8$, $a = 2$, so $y(1) = \boxed{-8}$ and the vertex is $(1, -8)$, a minimum.

Step 2. For the x -intercepts set $y = 0$: $2(x - 1)^2 = 8$, so $(x - 1)^2 = 4$ and $x - 1 = \pm 2$.

Step 3. x -intercepts $\boxed{x = -1 \text{ and } x = 3}$

Step 4. Since $|a| = 2 > 1$, the parabola is *narrower* than $y = x^2$: one unit from the axis it has already risen 2 instead of 1.

Sketch. Axis $x = 1$, y -intercept $y(0) = -6$.

6. $y = -x^2 + 6x - 5$.

Step 1. Here $a = -1$, $b = 6$, $c = -5$. Axis of symmetry: $x = -\frac{6}{2(-1)} = 3$. Keeping the minus sign on a is the whole difficulty.

Step 2. $y(3) = -9 + 18 - 5 = 4$. Because $a < 0$ this point is a *maximum* and the parabola opens downward.

Step 3. Vertex $(3, 4)$, a maximum, axis $x = 3$.

Step 4. y -intercept: $y(0) = -5$

Sketch. $-x^2 + 6x - 5 = -(x-1)(x-5)$, so the x -intercepts are 1 and 5, symmetric about $x = 3$.

Common wrong answer: -32 — the minus sign on a was dropped in $-\frac{b}{2a}$, putting the axis at $x = -3$.

7. $y = x^2 - 9$.

Step 1. Set $y = 0$: $x^2 = 9$. This is a difference of squares, $(x-3)(x+3) = 0$.

Step 2. x -intercepts $x = -3$ and $x = 3$

Step 3. There is no x term, so $b = 0$ and the axis of symmetry is $x = -\frac{0}{2} = 0$ — the y -axis itself. That is exactly why the two intercepts are mirror images of each other.

Step 4. $y(0) = -9$, so the vertex is $(0, -9)$, a minimum (the parabola opens upward).

Sketch. Symmetric points $(\pm 2, -5)$ help fix the width.

8. $y = \frac{1}{2}(x+4)^2 - 2$.

Step 1. Vertex form with $h = -4$, $k = -2$ and $a = \frac{1}{2}$, so $y(-4) = -2$ and the vertex is $(-4, -2)$.

Step 2. Set $y = 0$: $\frac{1}{2}(x+4)^2 = 2$, so $(x+4)^2 = 4$ and $x+4 = \pm 2$.

Step 3. x -intercepts $x = -6$ and $x = -2$

Step 4. $a = \frac{1}{2} > 0$, so it opens upward, and since $|a| < 1$ it is *wider* than $y = x^2$: one unit from the axis it rises only $\frac{1}{2}$.

Sketch. Axis $x = -4$; y -intercept $y(0) = 8 - 2 = 6$.

9. $y = 2x^2 - 12x + 10$.

Step 1. Every coefficient is even, so factor the leading coefficient out first: $2(x^2 - 6x + 5) = 2(x-1)(x-5)$.

Step 2. A product is zero when a factor is zero, and the 2 can never be zero, so $x = 1$ or $x = 5$.

Step 3. x -intercepts $x = 1$ and $x = 5$

Step 4. Midpoint of the intercepts: $x = \frac{1+5}{2} = 3$. Check against the formula: $-\frac{-12}{2(2)} = 3$

✓. Then $y(3) = 18 - 36 + 10 = -8$.

Step 5. Vertex $(3, -8)$, opening upward since $a = 2 > 0$.

Sketch. y -intercept 10, symmetric partner $(6, 10)$.

10. Intercepts -1 and 5 , through $(2, 9)$. *The sketch and the explanation are open work, flagged for manual review; the model answer below is what full credit contains.*

Step 1. Knowing both x -intercepts fixes the factors, so the equation must look like $y = a(x + 1)(x - 5)$. Only a is unknown, and one more point is exactly enough to find it.

Step 2. Substitute $(2, 9)$: $9 = a(2 + 1)(2 - 5) = a(3)(-3) = -9a$, so $a = -1$ and $y = -(x + 1)(x - 5)$.

Step 3. Multiply out: $-(x^2 - 4x - 5) = -x^2 + 4x + 5$. Its x -intercepts are still $x = -1$ and $x = 5$

Step 4. The axis is midway between the intercepts, at $x = \frac{-1+5}{2} = 2$, and the vertex height is $y(2) = -4 + 8 + 5 = 9$

Step 5 (the sketch). Vertex $(2, 9)$, axis $x = 2$, opening downward because $a = -1 < 0$; y -intercept 5 .

Full credit: the equation $y = -(x + 1)(x - 5) = -x^2 + 4x + 5$, a labelled sketch, and a sentence saying that the point $(2, 9)$ sits *above* the x -axis between the two intercepts, which can only happen if the parabola arches over — forcing a to be negative.